

- Module : Data Analytics
- Lecturer : MOUKAFIH Younes
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Lab 3 – Introduction to the scikit-learn module

Objective:

scikit-learn is a Python module entirely dedicated to data processing. Datasets are represented by vectors in a feature space (variables or "features"). The data is stored in a matrix of size $m \times k$, where m represents the row vectors for objects characterized by k variables. Additionally, this matrix comes with a set of m labels (which can be strings or numbers), one for each data point, and a set of k variable names (also strings or numbers), one for each variable. Data with the same label belong to the same class.

You can use `help(function_name)` to get help about a Python function.

Iris Dataset Overview:

The iris dataset is a classic dataset in machine learning that is often used as a benchmark for classification problems. It was first introduced by Sir Ronald Fisher in 1936 and consists of 150 samples of iris flowers, with 50 samples of each of the 3 different species: setosa, versicolor, and virginica. Each sample has 4 features: sepal length, sepal width, petal length, and petal width.

A. Importing Libraries

1. Import scikit-learn:

In Python, most functions are included in libraries that must be imported to be used. To import scikit-learn, write:

```
from sklearn import *
```

2. Import the libraries numpy (scientific computation) and matplotlib.pyplot (for figures).

B. Manipulating a Dataset

1. Load the Iris dataset with the command:

```
iris = datasets.load_iris()
```

The variable `iris` is a dictionary that contains the data matrix (`iris.data`), a vector of class numbers (`target`), the names of the variables (`feature_names`), and the class names (`target_names`).

2. Display the keys of the dictionary
3. Display the first 4 rows of data matrix `iris.data`. And display the target `iris.target`.
4. Display the feature names, and the target names (use `print`).
5. Display the class names for each data point. Store this array in a variable named '`species`'.

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iris.data is a matrix but also an object, like any variable in Python, which includes methods. For example, iris.data.mean(0) gives the mean of each variable, and iris.data.mean(1) gives the mean of each data point. There are also attributes associated with objects. For example, iris.target.size gives the number of objects in target.

6. Display the mean, standard deviation (std), minimum, and maximum for each variable.
7. Create a dataframe from the iris data matrix:

```
df = pd.DataFrame(data = iris.data, columns = iris.feature_names)
```

8. Visualize histograms for all the features in the dataset, set the numbers of bins to 12. Analyze the plots.
9. Calculate the correlation between the variables and visualize the correlation matrix using a heatmap (using seaborn). Interpret results.
10. Add the species array you created in question (5) to the dataframe as a new column.
11. Create a bar plot to show the frequency of each species in the dataset. Is this a balanced dataset?
12. Using seaborn, create side-by-side boxplots to visualize the distribution of 'petal length (cm)' for each of the three iris species.

End of the lab.